IQRA UNIVERSITY IU

FACULTY OF ENGINEERING SCIENCES AND TECHNOLOGY

Department: **Software Engineering** Program: **BS (SE)**

PROBABILITY & STATISTICS

Announced date: 06/05/2026 Due Date: 10/06/2026 Total Marks = 05

Complex Computing Problem (CCP)		
Mapped CLO	**SDG**	**Complex Problem Solving Mapped**
CLO3	4 & 9	WP1 (Depth of knowledge required) WP2 (Range of conflicting requirements required) WP3 (Depth of analysis required)

Title: Smart Technology Usage and Productivity Analysis System

Problem Statement

A technology research organization is conducting a study to analyze how digital technology usage affects student productivity and cognitive performance.

The organization wants to investigate whether certain measurable technology usage indicators can be used to predict academic productivity and performance efficiency.

For this purpose, data is collected from a sample of students, including:

- Daily Screen Time (hours)
- Social Media Usage (hours/day)
- Online Study Time (hours/day)
- Sleep Duration (hours/night)
- Device Type (Smartphone/Laptop/Tablet)
- Number of Apps Used Daily
- Productivity Score (out of 10)

The objective is to analyze relationships between technology usage patterns and productivity outcomes and identify which factors have the strongest impact on student performance.

Dataset

Student	Screen Time	Social Media	Study Time	Sleep	Device	Apps Used	Productivity Score
1	7	4	2	5	Smartphone	15	5
2	4	2	5	7	Laptop	10	8
3	8	5	1	4	Smartphone	18	4
4	3	1	6	8	Laptop	8	9
5	6	3	3	6	Tablet	12	6
6	9	6	1	4	Smartphone	20	3
7	5	2	4	7	Laptop	11	7
8	7	4	2	5	Smartphone	16	5
9	4	1	6	8	Laptop	9	9
10	6	3	3	6	Tablet	13	6

Deliverables

Part A: Problem Identification and Decision-Making Context

Identify and clearly articulate the decision-making problem presented in the given technology usage scenario.

Explain how multiple digital behavior indicators (screen time, social media usage, study time, sleep duration, device type, etc.) contribute to uncertainty in evaluating productivity.

Your response must include:

- Definition of the core decision-making problem
- Identification of conflicting technology usage variables
- Explanation of why statistical analysis is required for decision-making

Part B: Analytical Solution of the Problem

Provide a complete statistical and analytical solution to the problem identified in Part A.

Your solution must include:

- Descriptive statistics (mean, variance, standard deviation)
- Correlation analysis between technology usage factors and productivity score
- Probability-based interpretation of productivity levels
- Identification of key influencing digital behavior factors
- Preliminary predictive interpretation of productivity outcomes

Part C: Critical Evaluation of the Solution

Discuss and evaluate the solution provided in Part B under a modified scenario:

Assume that a key influencing variable (e.g., Social Media Usage) is removed from the dataset.

Analyze whether the relationship between remaining variables (screen time, study time, sleep duration, productivity score) still shows a significant association.

Your discussion must address:

- Impact of removing a key digital behavior variable on correlation results
- Reliability of productivity prediction after removal
- Changes in strength of association among remaining variables
- Interpretation of dependency among variables after exclusion

Part D: Alternative Approaches to Problem Solving

Critically analyze whether the problem and its solution can be solved using alternative methods.

Possible alternative approaches:

- Regression-based modeling instead of correlation analysis
- Machine learning approaches (decision trees, random forest, logistic regression)
- Weighted digital productivity index model
- Multi-criteria decision analysis (MCDA)

Explain how these methods:

- Improve prediction accuracy

- Handle multiple digital behavior variables effectively
- Provide deeper insights into productivity patterns

Category of Deliverables

Category	Deliverables
Conceptual	Problem statement, UML diagrams, flowcharts
Implementation	SPSS, Excel, Python or any other statistical/analytical tool
Report	Project report, user manual
Presentation	PowerPoint slides and Word/PDF report

CCP Attributes Mapped

WP	Description	Justification
WP1	Depth of Knowledge	Requires statistical modelling of technology usage data
WP2	Range of Conflicting Requirements	Requires managing conflicting effects of digital habits
WP3	Depth of Analysis	Involves correlation, probability, and predictive analysis